

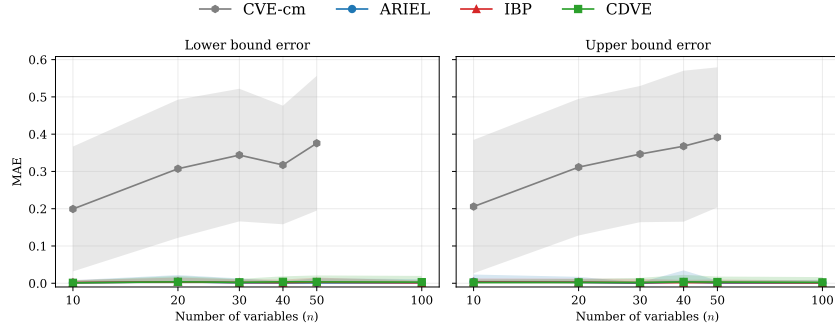


Figure 1: Large random dag LCNs (loopy, treewidth ≤ 3): MAE against the CredalJT reference, with bands giving one standard deviation over the ten instances per size and curves stopping where the budget was exceeded. Timings and the other three families are in Appendix 2.

1 Experiments

Benchmarks. We evaluate on two groups of instances. *Random LCNs* come in four families, each generated so that every sentence is *family-realizable*: `tree`, `polytree`, `ktree` ($k=3$) and `dag` (moralized treewidth ≤ 3 , at most two parents per atom). Family-realizability means each sentence constrains a single family’s conditional table alone; with the `linear` factorization the credal-network relaxation is exact *iff* the primal graph is singly connected *and* every sentence is realizable. So `tree` and `polytree` probe the tight regime, while `ktree` and `dag` are realizable but loopy — any two-parent collider moralizes into a cycle — and there the methods give outer bounds only. Each family is generated at two scales, *small* ($n \in \{5, \dots, 10\}$) and *large* ($n \in \{10, 20, 30, 40, 50, 100\}$), with 10 instances per size, giving $8 \times 60 = 480$ instances. Sentence bounds are drawn as $[\max(0.01, v - \varepsilon), \min(1, v + \varepsilon)]$ with $v \sim U(0, 1)$ and $\varepsilon = 0.3$; formulas span

Instance	Algorithm	MAE _l	MAE _u	w	Time (s)
Alarm ($n=10$), ref 0.72s	ARIEL	0.000	0.000	1.00	3.92
	IBP	0.101	0.015	1.37	0.13
	CDVE	0.101	0.015	1.37	0.69
	CVE _{cm}	0.087	0.067	0.86	0.24
Credit ($n=11$), ref 1.15s	ARIEL	0.050	0.000	1.18	3.18
	IBP	0.257	0.000	1.92	0.24
	CDVE	0.257	0.000	1.92	0.69
	CVE _{cm}	0.108	0.165	0.78	1.73
Hepatitis ($n=10$), ref 0.56s	ARIEL	0.008	0.008	1.02	3.91
	IBP	0.069	0.080	1.86	0.75
	CDVE	0.070	0.085	1.85	1.15
	CVE _{cm}	0.114	0.184	1.49	16.53
MScaling ($n=12$), ref 0.66s	ARIEL	0.028	0.000	1.73	8.13
	IBP	0.129	0.493	7.64	0.35
	CDVE	0.129	0.501	7.72	1.23
	CVE _{cm}	0.079	0.227	2.99	0.57
BScaling ($n=32$), ref 2.42s	ARIEL	0.000	0.000	1.00	19.80
	IBP	0.115	0.225	4.49	0.69
	CDVE	0.115	0.225	4.49	20.55
	CVE _{cm}	0.064	0.072	1.55	9.18
PScaling ($n=40$), ref 3.27s	ARIEL	0.000	0.000	1.00	34.72
	IBP	0.129	0.263	4.43	0.91
	CDVE	0.129	0.263	4.43	51.25
	CVE _{cm}	0.070	0.057	1.20	658.84

Table 1: Realistic LCNs: three instances derived from real Bayesian networks (top) and three Supreme Court chain-graph instances (bottom), all against the CredalJT reference. w is the mean interval-width ratio (> 1 looser than the reference, < 1 tighter). Time is total time including the credal-network build; the reference’s own time on each instance is given with the instance name.

at most three atoms; two extra marginals per instance are placed on root atoms only; the seed is 42. Every instance is accepted only after a product-distribution consistency witness is found (scope-local for $n > 10$, avoiding the 2^n term), so all 480 are consistent by construction. *Realistic LCNs* are of two kinds: ten instances derived from real Bayesian networks [1] ($n = 4$ –11, 6–24 sentences), and sixteen instances from the causal chain graphs of the US Supreme Court analysis of [2]. The latter stress two different axes: the `scaling` instances are large and sparse ($n = 6$ –40, 11–79 sentences), whereas the `supreme_court` instances are small but extremely dense ($n=10$ with 345–1009 sentences).

Algorithms, reference and metrics. We compare four approximate algorithms: **ARIEL** [3], which operates directly on the LCN factor graph (10 iterations) and needs no credal-network build; **IBP**, interval belief propagation on the credal network (100 iterations); **CDVE**, coordinate descent over extreme points (50 iterations), which by construction returns *inner* bounds; and **CVE_{cm}**, Credal Variable Elimination with its intermediate potentials clustered to at most 10 functions using the mean representative. Plain **CVE** (exact over the strong extension, no clustering) is included where it terminates. All credal-network methods share the same build: the `linear` factorization, no scope merging, and per-family local credal sets solved as linear programs. As the reference we use **CredalJT**, the junction-tree NLP formulation (`ipopt` backend, falling back to the certified SCIP one): it is the tightest engine tractable at every size here, whereas exact inference over the 2^n joint solves only 15/60 of `tree_small` and none of the Supreme Court instances. For every atom we report the mean absolute error of the lower and upper marginal bounds against the reference, $\text{MAE}_l = \frac{1}{n} \sum_i |P^*(x_i) - \underline{P}(x_i)|$ and MAE_u analogously, the mean interval-width ratio w (the algorithm’s width divided by the reference’s: $w > 1$ is looser, $w < 1$ tighter), and wall-clock time in seconds including the credal-network build. Note that CredalJT’s build is not comparable to the others’: it needs only the chain-graph structure and skips the vertex enumeration entirely.

Setup. All algorithms are implemented in Python 3.12 and use `ipopt` 3.14 [4] for the local programs and SCIP for the certified global ones. Every run is single-threaded with a limit of 3600 seconds and 30 GB of memory; a dash in a table means the algorithm did not finish inside that budget.

Random LCNs. Figure 1 shows the loopy `dag` family at the large scale, with shaded bands giving one standard deviation over the ten instances per size; the other three families are in Appendix 2. Three findings hold across all four families. First, **IBP dominates**: its error against the reference is at or below 4×10^{-3} on every family and size — indistinguishable from CredalJT on `tree` and `polytree`, as the exactness characterization predicts — while being the fastest method everywhere, by one to three orders of magnitude at $n=100$. Second, **CDVE matches that accuracy but not that cost**: its mean error stays in the 10^{-3} range, yet its runtime grows from under a second on the small families to 273–614 seconds on the large ones, making it slower than ARIEL at $n=100$. ARIEL is likewise accurate (exactly zero error on `tree` and `polytree`) but pays 25–35 seconds at scale. Third, **clustering buys speed at a real cost in quality**: **CVE_{cm}** runs in a fraction of a second on the small families but its error grows with n , from 0.08–0.11 on the small families to 0.31–0.35 on the large ones, and its width ratio of 0.73–0.86 shows the intervals are systematically *too tight* — the mean cluster representative is an interior point and does not preserve the enclosure. Plain **CVE** is the scalability bottleneck: it is essentially exact where it runs but costs 226–1141 seconds, times out on all of `ktree_small` and `polytree_large`, and is intractable on the large loopy families.

Realistic LCNs. Table 1 reports six representative instances. The ordering is different here: ARIEL is clearly the tightest method, at or near zero error on four of the six instances, because it reasons on the original LCN rather than on its credal-network relaxation. IBP and CDVE agree closely and produce widths 1.4 – $7.7 \times$ the reference — genuine but loose outer bounds, the gap being the cost of the relaxation rather than a defect of the message passing. On running time the picture is again favourable to IBP, which is the fastest method on every instance and the only one that stays under a second at $n=40$, where CDVE needs 51 seconds and **CVE_{cm}** 659. The nine dense `supreme_court` instances are omitted because only ARIEL and the reference complete on them: at $n=10$ with up to 1009 sentences it is the constraint system, not the treewidth, that defeats these methods.

64 2 Full Experimental Results

65 This appendix reports every measurement behind Section 1: all four random families at both scales,
66 and the complete per-instance results for the two realistic suites. Throughout, the reference is
67 CredalJT and the budget is 3600 seconds and 30 GB per run.

68 2.1 Coverage and exclusions

69 Not every algorithm completes on every instance, and reading the figures without this in mind would
70 overstate coverage, since a curve simply stops where its algorithm ran out of budget. Over the 480
71 random instances, ARIEL and IBP complete essentially all of them, and CredalJT's `ipopt` backend
72 completes all of the instances it was run on. The exceptions are concentrated in the CVE family.
73 Plain CVE, without clustering, is the least scalable: it solves 35/60 of `tree_small`, 30/60 of
74 `polytree_small` and 24/60 of `dag_small`, but *none* of `ktree_small` (21 timeouts and 39
75 errors) and none of `polytree_large`; on the other three large families it was not run at all. It
76 is therefore absent from most plots, and where present it covers only the smallest sizes. Clustering
77 restores coverage at small scale — CVE_{cm} and CVE_{cp} solve all 60 instances of every small family
78 — but degrades on the large loopy ones, where CVE_{cm} reaches only 14/60 on `ktree_large` and
79 30/60 on `dag_large`. Exact inference over the 2^n joint is viable only at the very smallest sizes
80 (15/60 on `tree_small`, 13/60 on `polytree_small`, 2/10 on the Bayesian-network instances,
81 none of the Supreme Court ones), which is why it is not used as the reference. Three bookkeeping
82 gaps are worth recording: the `ktree_large` CDVE log holds 58 records rather than 60, the
83 Supreme Court exact-inference log holds 12 rather than 16, and the Bayesian-network CVE log holds
84 7 rather than 10; the corresponding instances are simply missing rather than failed, and are excluded
85 from the aggregates.

86 2.2 Random LCNs

87 Figures 2–5 give the error and timing curves for all four families at both scales. The tables that follow
88 them give the underlying per-size aggregates; for each family and scale the first table reports means
89 and the second the corresponding standard deviations over the ten instances at each size.

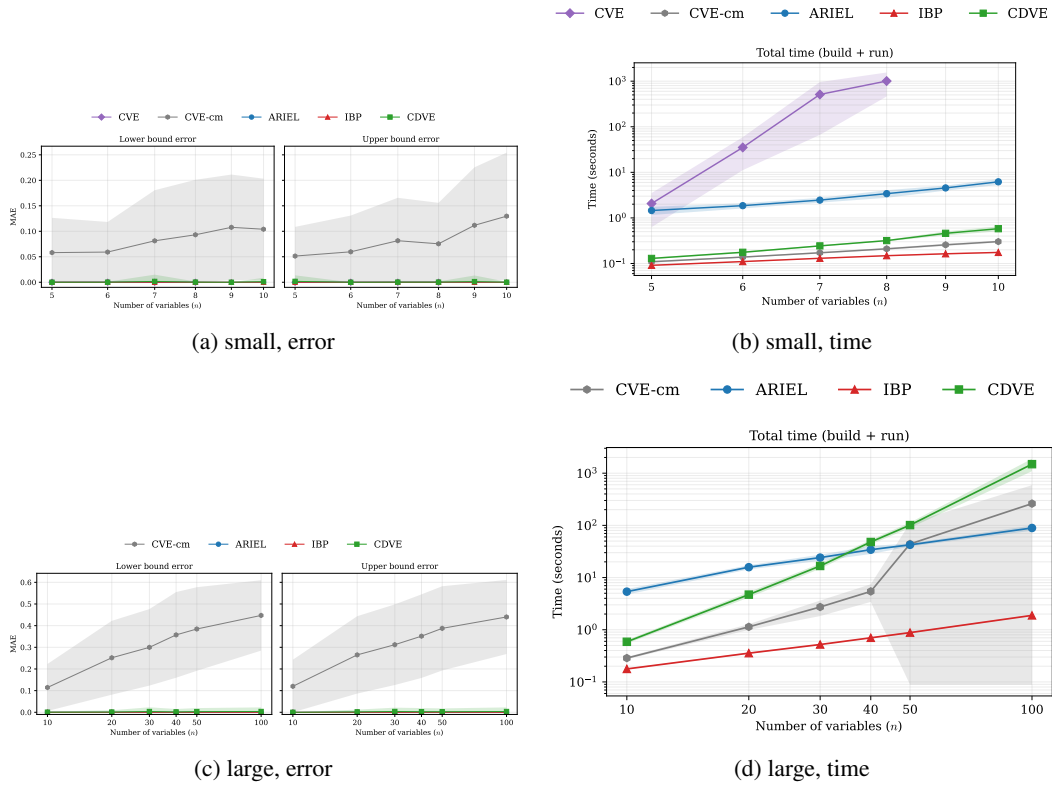


Figure 2: Random tree LCNs. Singly connected and fully realizable, so the credal-network relaxation is tight: ARIEL and IBP are indistinguishable from the reference.

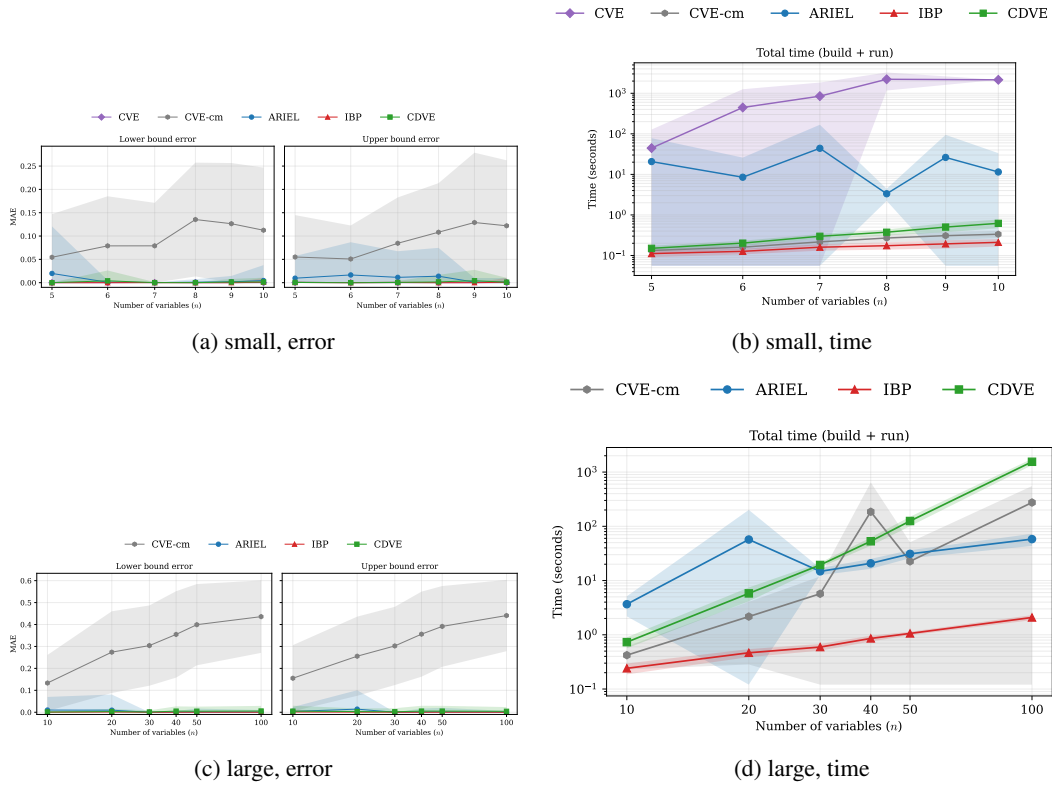


Figure 3: Random `polytree` LCNs. Also singly connected; the residual ARIEL error at the small scale comes from instances where the reference itself is the tighter engine.

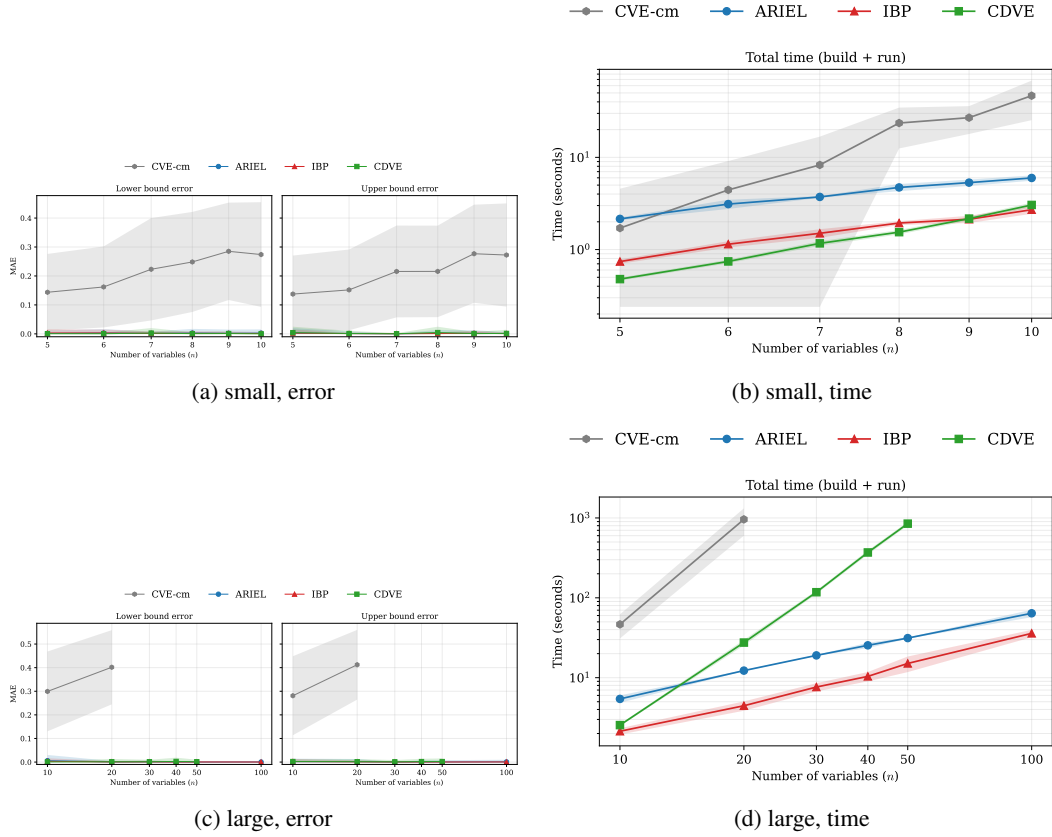


Figure 4: Random `ktree` LCNs ($k=3$). Loopy despite bounded treewidth, so all credal-network methods give outer bounds only; plain CVE solves none of these instances and CVE_{cm} stops after $n=20$.

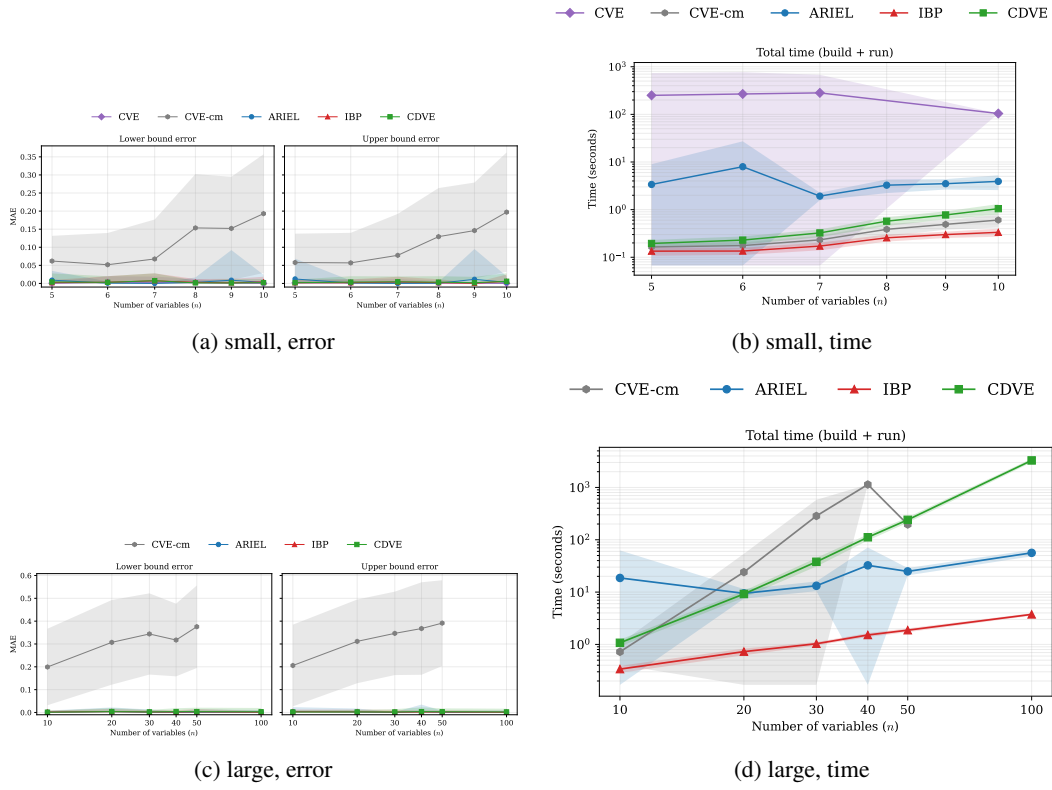


Figure 5: Random dag LCNs (moralized treewidth ≤ 3 , at most two parents). The least structured family; CDVE's runtime grows fastest here, exceeding 600 seconds at $n=100$.

Table 2: Mean error metrics vs. CJT reference

Type	n	Algorithm	MAE _{lb}	MAE _{ub}	Build	Run	Total	Ref	IW
tree_fr	5	CDVE	0.0001	0.0017	0.08	0.04	0.13	0.31	1.00
tree_fr	5	ARIEL	0.0000	0.0000	0.00	1.45	1.45	0.31	–
tree_fr	5	CVE	0.0000	0.0000	0.10	1.97	2.07	0.31	1.00
tree_fr	5	CVE _{cm}	0.0581	0.0514	0.09	0.02	0.11	0.31	1.00
tree_fr	5	IBP	0.0000	0.0000	0.09	0.00	0.09	0.31	1.00
tree_fr	6	CDVE	0.0001	0.0000	0.10	0.08	0.18	0.28	1.00
tree_fr	6	ARIEL	0.0000	0.0000	0.00	1.86	1.86	0.28	–
tree_fr	6	CVE	0.0000	0.0000	0.12	35.01	35.13	0.28	1.00
tree_fr	6	CVE _{cm}	0.0592	0.0597	0.10	0.04	0.14	0.28	1.00
tree_fr	6	IBP	0.0000	0.0000	0.11	0.00	0.11	0.28	1.00
tree_fr	7	CDVE	0.0016	0.0002	0.12	0.12	0.24	0.33	1.00
tree_fr	7	ARIEL	0.0000	0.0000	0.00	2.46	2.46	0.33	–
tree_fr	7	CVE	0.0000	0.0000	0.12	511.13	511.25	0.33	1.00
tree_fr	7	CVE _{cm}	0.0813	0.0815	0.12	0.05	0.17	0.33	1.00
tree_fr	7	IBP	0.0000	0.0000	0.13	0.00	0.13	0.33	1.00
tree_fr	8	CDVE	0.0002	0.0002	0.14	0.18	0.32	0.39	1.00
tree_fr	8	ARIEL	0.0000	0.0000	0.00	3.41	3.41	0.39	–
tree_fr	8	CVE	0.0000	0.0000	0.14	1006.64	1006.78	0.39	1.00
tree_fr	8	CVE _{cm}	0.0931	0.0754	0.14	0.07	0.21	0.39	1.00
tree_fr	8	IBP	0.0000	0.0000	0.14	0.01	0.15	0.39	1.00
tree_fr	9	CDVE	0.0000	0.0013	0.15	0.31	0.46	0.44	1.00
tree_fr	9	ARIEL	0.0000	0.0000	0.00	4.55	4.55	0.44	–
tree_fr	9	CVE _{cm}	0.1078	0.1117	0.16	0.10	0.26	0.44	1.00
tree_fr	9	IBP	0.0000	0.0000	0.16	0.01	0.16	0.44	1.00
tree_fr	10	CDVE	0.0011	0.0001	0.17	0.41	0.58	0.50	1.00
tree_fr	10	ARIEL	0.0000	0.0000	0.00	6.19	6.19	0.50	–
tree_fr	10	CVE _{cm}	0.1040	0.1297	0.18	0.12	0.30	0.50	1.00
tree_fr	10	IBP	0.0000	0.0000	0.17	0.01	0.17	0.50	1.00

Table 3: Standard deviation of error metrics vs. CJT reference

Type	n	Algorithm	SD _{lb}	SD _{ub}	Build	Run	Total	Ref	IW
tree_fr	5	CDVE	0.0005	0.0119	0.08	0.04	0.13	0.31	1.00
tree_fr	5	ARIEL	0.0000	0.0000	0.00	1.45	1.45	0.31	–
tree_fr	5	CVE	0.0000	0.0000	0.10	1.97	2.07	0.31	1.00
tree_fr	5	CVE _{cm}	0.0683	0.0572	0.09	0.02	0.11	0.31	1.00
tree_fr	5	IBP	0.0000	0.0000	0.09	0.00	0.09	0.31	1.00
tree_fr	6	CDVE	0.0009	0.0000	0.10	0.08	0.18	0.28	1.00
tree_fr	6	ARIEL	0.0000	0.0000	0.00	1.86	1.86	0.28	–
tree_fr	6	CVE	0.0000	0.0000	0.12	35.01	35.13	0.28	1.00
tree_fr	6	CVE _{cm}	0.0591	0.0711	0.10	0.04	0.14	0.28	1.00
tree_fr	6	IBP	0.0000	0.0000	0.11	0.00	0.11	0.28	1.00
tree_fr	7	CDVE	0.0133	0.0014	0.12	0.12	0.24	0.33	1.00
tree_fr	7	ARIEL	0.0000	0.0000	0.00	2.46	2.46	0.33	–
tree_fr	7	CVE	0.0000	0.0000	0.12	511.13	511.25	0.33	1.00
tree_fr	7	CVE _{cm}	0.0993	0.0842	0.12	0.05	0.17	0.33	1.00
tree_fr	7	IBP	0.0000	0.0000	0.13	0.00	0.13	0.33	1.00
tree_fr	8	CDVE	0.0013	0.0014	0.14	0.18	0.32	0.39	1.00
tree_fr	8	ARIEL	0.0000	0.0000	0.00	3.41	3.41	0.39	–
tree_fr	8	CVE	0.0000	0.0000	0.14	1006.64	1006.78	0.39	1.00
tree_fr	8	CVE _{cm}	0.1078	0.0804	0.14	0.07	0.21	0.39	1.00
tree_fr	8	IBP	0.0000	0.0000	0.14	0.01	0.15	0.39	1.00
tree_fr	9	CDVE	0.0000	0.0120	0.15	0.31	0.46	0.44	1.00
tree_fr	9	ARIEL	0.0000	0.0000	0.00	4.55	4.55	0.44	–
tree_fr	9	CVE _{cm}	0.1035	0.1138	0.16	0.10	0.26	0.44	1.00
tree_fr	9	IBP	0.0000	0.0000	0.16	0.01	0.16	0.44	1.00
tree_fr	10	CDVE	0.0076	0.0009	0.17	0.41	0.58	0.50	1.00
tree_fr	10	ARIEL	0.0000	0.0000	0.00	6.19	6.19	0.50	–
tree_fr	10	CVE _{cm}	0.0992	0.1246	0.18	0.12	0.30	0.50	1.00
tree_fr	10	IBP	0.0000	0.0000	0.17	0.01	0.17	0.50	1.00

Table 4: Mean error metrics vs. CJT reference

Type	n	Algorithm	MAE _{lb}	MAE _{ub}	Build	Run	Total	Ref	IW
tree_fr	10	CDVE	0.0005	0.0000	0.18	0.41	0.59	0.64	1.00
tree_fr	10	ARIEL	0.0000	0.0000	0.00	5.37	5.37	0.64	–
tree_fr	10	CVE _{cm}	0.1145	0.1199	0.18	0.11	0.29	0.64	1.00
tree_fr	10	IBP	0.0000	0.0000	0.17	0.01	0.18	0.64	1.00
tree_fr	20	CDVE	0.0011	0.0019	0.35	4.36	4.71	1.43	1.00
tree_fr	20	ARIEL	0.0000	0.0000	0.00	15.80	15.80	1.43	–
tree_fr	20	CVE _{cm}	0.2517	0.2650	0.37	0.77	1.14	1.43	1.00
tree_fr	20	IBP	0.0000	0.0000	0.34	0.02	0.36	1.43	1.00
tree_fr	30	CDVE	0.0036	0.0037	0.50	16.18	16.68	2.58	1.00
tree_fr	30	ARIEL	0.0000	0.0000	0.00	24.13	24.13	2.58	–
tree_fr	30	CVE _{cm}	0.2998	0.3120	0.55	2.18	2.73	2.58	1.00
tree_fr	30	IBP	0.0000	0.0000	0.49	0.03	0.52	2.58	1.00
tree_fr	40	CDVE	0.0017	0.0026	0.66	47.39	48.04	3.90	1.00
tree_fr	40	ARIEL	0.0000	0.0000	0.00	34.21	34.21	3.90	–
tree_fr	40	CVE _{cm}	0.3573	0.3515	0.70	4.72	5.43	3.90	1.00
tree_fr	40	IBP	0.0000	0.0000	0.66	0.04	0.70	3.90	1.00
tree_fr	50	CDVE	0.0031	0.0026	0.83	100.39	101.22	5.95	1.00
tree_fr	50	ARIEL	0.0000	0.0000	0.00	42.41	42.41	5.95	–
tree_fr	50	CVE _{cm}	0.3845	0.3875	0.96	43.09	44.05	5.95	1.00
tree_fr	50	IBP	0.0000	0.0000	0.83	0.05	0.88	5.95	1.00
tree_fr	100	CDVE	0.0037	0.0036	1.67	1487.61	1489.29	16.24	1.00
tree_fr	100	ARIEL	0.0000	0.0000	0.00	89.09	89.09	16.24	–
tree_fr	100	CVE _{cm}	0.4473	0.4401	1.94	259.37	261.31	16.24	1.00
tree_fr	100	IBP	0.0000	0.0000	1.76	0.12	1.88	16.24	1.00

Table 5: Standard deviation of error metrics vs. CJT reference

Type	n	Algorithm	SD _{lb}	SD _{ub}	Build	Run	Total	Ref	IW
tree_fr	10	CDVE	0.0026	0.0003	0.18	0.41	0.59	0.64	1.00
tree_fr	10	ARIEL	0.0000	0.0000	0.00	5.37	5.37	0.64	–
tree_fr	10	CVE _{cm}	0.1085	0.1227	0.18	0.11	0.29	0.64	1.00
tree_fr	10	IBP	0.0000	0.0000	0.17	0.01	0.18	0.64	1.00
tree_fr	20	CDVE	0.0083	0.0104	0.35	4.36	4.71	1.43	1.00
tree_fr	20	ARIEL	0.0000	0.0000	0.00	15.80	15.80	1.43	–
tree_fr	20	CVE _{cm}	0.1704	0.1786	0.37	0.77	1.14	1.43	1.00
tree_fr	20	IBP	0.0000	0.0000	0.34	0.02	0.36	1.43	1.00
tree_fr	30	CDVE	0.0195	0.0177	0.50	16.18	16.68	2.58	1.00
tree_fr	30	ARIEL	0.0000	0.0000	0.00	24.13	24.13	2.58	–
tree_fr	30	CVE _{cm}	0.1766	0.1860	0.55	2.18	2.73	2.58	1.00
tree_fr	30	IBP	0.0000	0.0000	0.49	0.03	0.52	2.58	1.00
tree_fr	40	CDVE	0.0134	0.0169	0.66	47.39	48.04	3.90	1.00
tree_fr	40	ARIEL	0.0000	0.0000	0.00	34.21	34.21	3.90	–
tree_fr	40	CVE _{cm}	0.1977	0.1929	0.70	4.72	5.43	3.90	1.00
tree_fr	40	IBP	0.0000	0.0000	0.66	0.04	0.70	3.90	1.00
tree_fr	50	CDVE	0.0164	0.0154	0.83	100.39	101.22	5.95	1.00
tree_fr	50	ARIEL	0.0000	0.0000	0.00	42.41	42.41	5.95	–
tree_fr	50	CVE _{cm}	0.1929	0.1944	0.96	43.09	44.05	5.95	1.00
tree_fr	50	IBP	0.0000	0.0000	0.83	0.05	0.88	5.95	1.00
tree_fr	100	CDVE	0.0189	0.0190	1.67	1487.61	1489.29	16.24	1.00
tree_fr	100	ARIEL	0.0000	0.0000	0.00	89.09	89.09	16.24	–
tree_fr	100	CVE _{cm}	0.1625	0.1711	1.94	259.37	261.31	16.24	1.00
tree_fr	100	IBP	0.0000	0.0000	1.76	0.12	1.88	16.24	1.00

Table 6: Mean error metrics vs. CJT reference

Type	n	Algorithm	MAE _{lb}	MAE _{ub}	Build	Run	Total	Ref	IW
polytree_fr	5	CDVE	0.0001	0.0005	0.10	0.05	0.15	0.39	1.50
polytree_fr	5	ARIEL	0.0198	0.0096	0.00	20.68	20.68	0.39	–
polytree_fr	5	CVE	0.0000	0.0005	0.11	44.41	44.52	0.39	1.50
polytree_fr	5	CVE _{cm}	0.0545	0.0550	0.11	0.03	0.13	0.39	1.50
polytree_fr	5	IBP	0.0000	0.0005	0.11	0.01	0.11	0.39	1.50
polytree_fr	6	CDVE	0.0039	0.0000	0.12	0.08	0.20	0.44	1.40
polytree_fr	6	ARIEL	0.0010	0.0166	0.00	8.51	8.51	0.44	–
polytree_fr	6	CVE	0.0000	0.0000	0.14	445.58	445.71	0.44	1.40
polytree_fr	6	CVE _{cm}	0.0789	0.0508	0.12	0.04	0.16	0.44	1.40
polytree_fr	6	IBP	0.0000	0.0000	0.12	0.01	0.13	0.44	1.40
polytree_fr	7	CDVE	0.0002	0.0004	0.15	0.15	0.30	0.73	1.70
polytree_fr	7	ARIEL	0.0002	0.0114	0.00	44.04	44.04	0.73	–
polytree_fr	7	CVE	0.0003	0.0000	0.13	849.79	849.92	0.73	1.40
polytree_fr	7	CVE _{cm}	0.0790	0.0845	0.15	0.06	0.22	0.73	1.70
polytree_fr	7	IBP	0.0002	0.0004	0.15	0.01	0.16	0.73	1.70
polytree_fr	8	CDVE	0.0001	0.0017	0.16	0.22	0.38	0.91	1.40
polytree_fr	8	ARIEL	0.0010	0.0139	0.00	3.33	3.33	0.91	–
polytree_fr	8	CVE	0.0000	0.0000	0.15	2213.21	2213.37	0.91	1.00
polytree_fr	8	CVE _{cm}	0.1354	0.1082	0.18	0.09	0.27	0.91	1.40
polytree_fr	8	IBP	0.0000	0.0000	0.17	0.01	0.17	0.91	1.40
polytree_fr	9	CDVE	0.0015	0.0042	0.18	0.32	0.50	0.99	1.30
polytree_fr	9	ARIEL	0.0019	0.0012	0.00	26.20	26.20	0.99	–
polytree_fr	9	CVE _{cm}	0.1264	0.1290	0.19	0.12	0.31	0.99	1.30
polytree_fr	9	IBP	0.0000	0.0000	0.18	0.01	0.19	0.99	1.30
polytree_fr	10	CDVE	0.0014	0.0011	0.19	0.43	0.62	1.11	1.30
polytree_fr	10	ARIEL	0.0046	0.0011	0.00	11.47	11.47	1.11	–
polytree_fr	10	CVE	0.0000	0.0000	0.16	2149.12	2149.28	1.11	1.00
polytree_fr	10	CVE _{cm}	0.1124	0.1220	0.20	0.14	0.33	1.11	1.30
polytree_fr	10	IBP	0.0010	0.0010	0.20	0.01	0.21	1.11	1.30

Table 7: Standard deviation of error metrics vs. CJT reference

Type	n	Algorithm	SD _{lb}	SD _{ub}	Build	Run	Total	Ref	IW
polytree_fr	5	CDVE	0.0004	0.0029	0.10	0.05	0.15	0.39	1.50
polytree_fr	5	ARIEL	0.1010	0.0475	0.00	20.68	20.68	0.39	–
polytree_fr	5	CVE	0.0001	0.0029	0.11	44.41	44.52	0.39	1.50
polytree_fr	5	CVE _{cm}	0.0925	0.0898	0.11	0.03	0.13	0.39	1.50
polytree_fr	5	IBP	0.0001	0.0029	0.11	0.01	0.11	0.39	1.50
polytree_fr	6	CDVE	0.0222	0.0000	0.12	0.08	0.20	0.44	1.40
polytree_fr	6	ARIEL	0.0042	0.0702	0.00	8.51	8.51	0.44	–
polytree_fr	6	CVE	0.0000	0.0000	0.14	445.58	445.71	0.44	1.40
polytree_fr	6	CVE _{cm}	0.1063	0.0718	0.12	0.04	0.16	0.44	1.40
polytree_fr	6	IBP	0.0000	0.0000	0.12	0.01	0.13	0.44	1.40
polytree_fr	7	CDVE	0.0012	0.0032	0.15	0.15	0.30	0.73	1.70
polytree_fr	7	ARIEL	0.0009	0.0560	0.00	44.04	44.04	0.73	–
polytree_fr	7	CVE	0.0017	0.0001	0.13	849.79	849.92	0.73	1.40
polytree_fr	7	CVE _{cm}	0.0922	0.0982	0.15	0.06	0.22	0.73	1.70
polytree_fr	7	IBP	0.0012	0.0032	0.15	0.01	0.16	0.73	1.70
polytree_fr	8	CDVE	0.0011	0.0150	0.16	0.22	0.38	0.91	1.40
polytree_fr	8	ARIEL	0.0064	0.0608	0.00	3.33	3.33	0.91	–
polytree_fr	8	CVE	0.0000	0.0000	0.15	2213.21	2213.37	0.91	1.00
polytree_fr	8	CVE _{cm}	0.1220	0.1051	0.18	0.09	0.27	0.91	1.40
polytree_fr	8	IBP	0.0000	0.0000	0.17	0.01	0.17	0.91	1.40
polytree_fr	9	CDVE	0.0067	0.0234	0.18	0.32	0.50	0.99	1.30
polytree_fr	9	ARIEL	0.0126	0.0088	0.00	26.20	26.20	0.99	–
polytree_fr	9	CVE _{cm}	0.1301	0.1497	0.19	0.12	0.31	0.99	1.30
polytree_fr	9	IBP	0.0000	0.0001	0.18	0.01	0.19	0.99	1.30
polytree_fr	10	CDVE	0.0093	0.0094	0.19	0.43	0.62	1.11	1.30
polytree_fr	10	ARIEL	0.0331	0.0088	0.00	11.47	11.47	1.11	–
polytree_fr	10	CVE	0.0000	0.0000	0.16	2149.12	2149.28	1.11	1.00
polytree_fr	10	CVE _{cm}	0.1343	0.1404	0.20	0.14	0.33	1.11	1.30
polytree_fr	10	IBP	0.0092	0.0094	0.20	0.01	0.21	1.11	1.30

Table 8: Mean error metrics vs. CJT reference

Type	n	Algorithm	MAE _{lb}	MAE _{ub}	Build	Run	Total	Ref	IW
polytree_fr	10	CDVE	0.0010	0.0050	0.23	0.51	0.73	13.77	1.50
polytree_fr	10	ARIEL	0.0100	0.0038	0.00	3.66	3.66	13.77	–
polytree_fr	10	CVE _{cm}	0.1333	0.1551	0.22	0.20	0.42	13.77	1.50
polytree_fr	10	IBP	0.0003	0.0013	0.23	0.01	0.24	13.77	1.50
polytree_fr	20	CDVE	0.0031	0.0031	0.45	5.35	5.80	3.21	1.60
polytree_fr	20	ARIEL	0.0095	0.0143	0.00	57.07	57.07	3.21	–
polytree_fr	20	CVE _{cm}	0.2740	0.2555	0.44	1.73	2.17	3.21	1.60
polytree_fr	20	IBP	0.0012	0.0006	0.43	0.03	0.47	3.21	1.60
polytree_fr	30	CDVE	0.0012	0.0023	0.54	18.84	19.38	5.22	1.10
polytree_fr	30	ARIEL	0.0000	0.0000	0.00	14.78	14.78	5.22	–
polytree_fr	30	CVE _{cm}	0.3039	0.3023	0.58	5.12	5.71	5.22	1.10
polytree_fr	30	IBP	0.0001	0.0003	0.56	0.04	0.59	5.22	1.10
polytree_fr	40	CDVE	0.0051	0.0052	0.76	52.22	52.98	12.10	1.40
polytree_fr	40	ARIEL	0.0007	0.0009	0.00	20.76	20.76	12.10	–
polytree_fr	40	CVE _{cm}	0.3548	0.3560	0.78	184.60	185.37	12.10	1.40
polytree_fr	40	IBP	0.0003	0.0002	0.80	0.06	0.86	12.10	1.40
polytree_fr	50	CDVE	0.0047	0.0051	1.10	124.13	125.23	38.34	1.50
polytree_fr	50	ARIEL	0.0008	0.0012	0.00	31.07	31.07	38.34	–
polytree_fr	50	CVE _{cm}	0.3992	0.3913	0.93	21.67	22.60	38.34	1.50
polytree_fr	50	IBP	0.0005	0.0003	0.99	0.07	1.06	38.34	1.50
polytree_fr	100	CDVE	0.0056	0.0043	1.95	1545.82	1547.77	68.64	1.40
polytree_fr	100	ARIEL	0.0005	0.0003	0.00	58.08	58.08	68.64	–
polytree_fr	100	CVE _{cm}	0.4359	0.4408	1.86	272.65	274.51	68.64	1.33
polytree_fr	100	IBP	0.0002	0.0002	1.93	0.15	2.08	68.64	1.40

Table 9: Standard deviation of error metrics vs. CJT reference

Type	n	Algorithm	SD _{lb}	SD _{ub}	Build	Run	Total	Ref	IW
polytree_fr	10	CDVE	0.0041	0.0251	0.23	0.51	0.73	13.77	1.50
polytree_fr	10	ARIEL	0.0598	0.0200	0.00	3.66	3.66	13.77	–
polytree_fr	10	CVE _{cm}	0.1278	0.1507	0.22	0.20	0.42	13.77	1.50
polytree_fr	10	IBP	0.0013	0.0105	0.23	0.01	0.24	13.77	1.50
polytree_fr	20	CDVE	0.0127	0.0143	0.45	5.35	5.80	3.21	1.60
polytree_fr	20	ARIEL	0.0714	0.0862	0.00	57.07	57.07	3.21	–
polytree_fr	20	CVE _{cm}	0.1862	0.1804	0.44	1.73	2.17	3.21	1.60
polytree_fr	20	IBP	0.0060	0.0043	0.43	0.03	0.47	3.21	1.60
polytree_fr	30	CDVE	0.0076	0.0161	0.54	18.84	19.38	5.22	1.10
polytree_fr	30	ARIEL	0.0000	0.0000	0.00	14.78	14.78	5.22	–
polytree_fr	30	CVE _{cm}	0.1829	0.1783	0.58	5.12	5.71	5.22	1.10
polytree_fr	30	IBP	0.0014	0.0027	0.56	0.04	0.59	5.22	1.10
polytree_fr	40	CDVE	0.0218	0.0244	0.76	52.22	52.98	12.10	1.40
polytree_fr	40	ARIEL	0.0052	0.0162	0.00	20.76	20.76	12.10	–
polytree_fr	40	CVE _{cm}	0.1969	0.1938	0.78	184.60	185.37	12.10	1.40
polytree_fr	40	IBP	0.0019	0.0019	0.80	0.06	0.86	12.10	1.40
polytree_fr	50	CDVE	0.0210	0.0242	1.10	124.13	125.23	38.34	1.50
polytree_fr	50	ARIEL	0.0160	0.0190	0.00	31.07	31.07	38.34	–
polytree_fr	50	CVE _{cm}	0.1852	0.1840	0.93	21.67	22.60	38.34	1.50
polytree_fr	50	IBP	0.0049	0.0022	0.99	0.07	1.06	38.34	1.50
polytree_fr	100	CDVE	0.0235	0.0195	1.95	1545.82	1547.77	68.64	1.40
polytree_fr	100	ARIEL	0.0100	0.0060	0.00	58.08	58.08	68.64	–
polytree_fr	100	CVE _{cm}	0.1651	0.1624	1.86	272.65	274.51	68.64	1.33
polytree_fr	100	IBP	0.0025	0.0025	1.93	0.15	2.08	68.64	1.40

Table 10: Mean error metrics vs. CJT reference

Type	n	Algorithm	MAE _{lb}	MAE _{ub}	Build	Run	Total	Ref	IW
ktree_fr_k3	5	CDVE	0.0006	0.0042	0.33	0.14	0.48	0.29	3.00
ktree_fr_k3	5	ARIEL	0.0015	0.0028	0.00	2.15	2.15	0.29	–
ktree_fr_k3	5	CVE _{cm}	0.1437	0.1374	0.33	1.38	1.71	0.29	3.00
ktree_fr_k3	5	IBP	0.0030	0.0027	0.33	0.41	0.74	0.29	3.00
ktree_fr_k3	6	CDVE	0.0006	0.0012	0.45	0.29	0.74	0.36	3.00
ktree_fr_k3	6	ARIEL	0.0029	0.0011	0.00	3.11	3.11	0.36	–
ktree_fr_k3	6	CVE _{cm}	0.1620	0.1519	0.45	3.99	4.43	0.36	3.00
ktree_fr_k3	6	IBP	0.0030	0.0008	0.45	0.69	1.14	0.36	3.00
ktree_fr_k3	7	CDVE	0.0027	0.0005	0.57	0.59	1.17	0.42	3.00
ktree_fr_k3	7	ARIEL	0.0015	0.0002	0.00	3.72	3.72	0.42	–
ktree_fr_k3	7	CVE _{cm}	0.2231	0.2155	0.60	7.68	8.27	0.42	3.00
ktree_fr_k3	7	IBP	0.0018	0.0005	0.58	0.92	1.50	0.42	3.00
ktree_fr_k3	8	CDVE	0.0008	0.0044	0.64	0.90	1.54	0.51	3.00
ktree_fr_k3	8	ARIEL	0.0028	0.0020	0.00	4.72	4.72	0.51	–
ktree_fr_k3	8	CVE _{cm}	0.2484	0.2158	0.69	22.88	23.57	0.51	3.00
ktree_fr_k3	8	IBP	0.0018	0.0011	0.71	1.23	1.94	0.51	3.00
ktree_fr_k3	9	CDVE	0.0011	0.0009	0.78	1.39	2.17	0.60	3.00
ktree_fr_k3	9	ARIEL	0.0019	0.0017	0.00	5.32	5.32	0.60	–
ktree_fr_k3	9	CVE _{cm}	0.2851	0.2768	0.82	26.13	26.95	0.60	3.00
ktree_fr_k3	9	IBP	0.0015	0.0020	0.76	1.38	2.13	0.60	3.00
ktree_fr_k3	10	CDVE	0.0002	0.0015	0.95	2.10	3.05	0.68	3.00
ktree_fr_k3	10	ARIEL	0.0027	0.0011	0.00	5.98	5.98	0.68	–
ktree_fr_k3	10	CVE _{cm}	0.2740	0.2723	0.96	45.69	46.65	0.68	3.00
ktree_fr_k3	10	IBP	0.0013	0.0009	0.97	1.74	2.70	0.68	3.00

Table 11: Standard deviation of error metrics vs. CJT reference

Type	n	Algorithm	SD _{lb}	SD _{ub}	Build	Run	Total	Ref	IW
ktree_fr_k3	5	CDVE	0.0021	0.0206	0.33	0.14	0.48	0.29	3.00
ktree_fr_k3	5	ARIEL	0.0105	0.0171	0.00	2.15	2.15	0.29	–
ktree_fr_k3	5	CVE _{cm}	0.1324	0.1330	0.33	1.38	1.71	0.29	3.00
ktree_fr_k3	5	IBP	0.0125	0.0122	0.33	0.41	0.74	0.29	3.00
ktree_fr_k3	6	CDVE	0.0034	0.0094	0.45	0.29	0.74	0.36	3.00
ktree_fr_k3	6	ARIEL	0.0137	0.0071	0.00	3.11	3.11	0.36	–
ktree_fr_k3	6	CVE _{cm}	0.1399	0.1394	0.45	3.99	4.43	0.36	3.00
ktree_fr_k3	6	IBP	0.0119	0.0037	0.45	0.69	1.14	0.36	3.00
ktree_fr_k3	7	CDVE	0.0174	0.0022	0.57	0.59	1.17	0.42	3.00
ktree_fr_k3	7	ARIEL	0.0063	0.0013	0.00	3.72	3.72	0.42	–
ktree_fr_k3	7	CVE _{cm}	0.1767	0.1585	0.60	7.68	8.27	0.42	3.00
ktree_fr_k3	7	IBP	0.0069	0.0027	0.58	0.92	1.50	0.42	3.00
ktree_fr_k3	8	CDVE	0.0050	0.0202	0.64	0.90	1.54	0.51	3.00
ktree_fr_k3	8	ARIEL	0.0144	0.0114	0.00	4.72	4.72	0.51	–
ktree_fr_k3	8	CVE _{cm}	0.1728	0.1580	0.69	22.88	23.57	0.51	3.00
ktree_fr_k3	8	IBP	0.0071	0.0041	0.71	1.23	1.94	0.51	3.00
ktree_fr_k3	9	CDVE	0.0067	0.0057	0.78	1.39	2.17	0.60	3.00
ktree_fr_k3	9	ARIEL	0.0131	0.0110	0.00	5.32	5.32	0.60	–
ktree_fr_k3	9	CVE _{cm}	0.1681	0.1694	0.82	26.13	26.95	0.60	3.00
ktree_fr_k3	9	IBP	0.0068	0.0089	0.76	1.38	2.13	0.60	3.00
ktree_fr_k3	10	CDVE	0.0012	0.0127	0.95	2.10	3.05	0.68	3.00
ktree_fr_k3	10	ARIEL	0.0129	0.0052	0.00	5.98	5.98	0.68	–
ktree_fr_k3	10	CVE _{cm}	0.1808	0.1783	0.96	45.69	46.65	0.68	3.00
ktree_fr_k3	10	IBP	0.0054	0.0039	0.97	1.74	2.70	0.68	3.00

Table 12: Mean error metrics vs. CJT reference

Type	n	Algorithm	MAE _{lb}	MAE _{ub}	Build	Run	Total	Ref	IW
ktree_fr_k3	10	CDVE	0.0007	0.0023	0.40	2.14	2.55	0.65	3.00
ktree_fr_k3	10	ARIEL	0.0060	0.0017	0.00	5.42	5.42	0.65	–
ktree_fr_k3	10	CVE _{cm}	0.2993	0.2810	0.40	46.11	46.51	0.65	3.00
ktree_fr_k3	10	IBP	0.0032	0.0021	0.40	1.74	2.13	0.65	3.00
ktree_fr_k3	20	CDVE	0.0013	0.0017	0.85	26.67	27.52	1.57	3.00
ktree_fr_k3	20	ARIEL	0.0006	0.0014	0.00	12.26	12.26	1.57	–
ktree_fr_k3	20	CVE _{cm}	0.4017	0.4124	0.81	959.08	959.88	1.57	3.00
ktree_fr_k3	20	IBP	0.0007	0.0009	0.83	3.62	4.45	1.57	3.00
ktree_fr_k3	30	CDVE	0.0013	0.0011	1.28	116.25	117.53	2.91	3.00
ktree_fr_k3	30	ARIEL	0.0009	0.0001	0.00	19.01	19.01	2.91	–
ktree_fr_k3	30	IBP	0.0009	0.0004	1.26	6.38	7.64	2.91	3.00
ktree_fr_k3	40	CDVE	0.0027	0.0020	1.71	367.85	369.55	4.80	3.00
ktree_fr_k3	40	ARIEL	0.0003	0.0004	0.00	25.41	25.41	4.80	–
ktree_fr_k3	40	IBP	0.0003	0.0003	1.68	8.70	10.38	4.80	3.00
ktree_fr_k3	50	CDVE	0.0015	0.0022	3.68	845.10	848.78	6.89	3.00
ktree_fr_k3	50	ARIEL	0.0001	0.0004	0.00	31.32	31.32	6.89	–
ktree_fr_k3	50	IBP	0.0004	0.0003	3.64	11.44	15.08	6.89	3.00
ktree_fr_k3	100	ARIEL	0.0003	0.0007	0.00	64.03	64.03	22.71	–
ktree_fr_k3	100	IBP	0.0002	0.0005	10.55	25.34	35.89	22.71	3.00

Table 13: Standard deviation of error metrics vs. CJT reference

Type	n	Algorithm	SD _{lb}	SD _{ub}	Build	Run	Total	Ref	IW
ktree_fr_k3	10	CDVE	0.0031	0.0095	0.40	2.14	2.55	0.65	3.00
ktree_fr_k3	10	ARIEL	0.0242	0.0128	0.00	5.42	5.42	0.65	–
ktree_fr_k3	10	CVE _{cm}	0.1690	0.1673	0.40	46.11	46.51	0.65	3.00
ktree_fr_k3	10	IBP	0.0128	0.0113	0.40	1.74	2.13	0.65	3.00
ktree_fr_k3	20	CDVE	0.0122	0.0111	0.85	26.67	27.52	1.57	3.00
ktree_fr_k3	20	ARIEL	0.0041	0.0116	0.00	12.26	12.26	1.57	–
ktree_fr_k3	20	CVE _{cm}	0.1572	0.1476	0.81	959.08	959.88	1.57	3.00
ktree_fr_k3	20	IBP	0.0038	0.0068	0.83	3.62	4.45	1.57	3.00
ktree_fr_k3	30	CDVE	0.0085	0.0075	1.28	116.25	117.53	2.91	3.00
ktree_fr_k3	30	ARIEL	0.0055	0.0011	0.00	19.01	19.01	2.91	–
ktree_fr_k3	30	IBP	0.0055	0.0032	1.26	6.38	7.64	2.91	3.00
ktree_fr_k3	40	CDVE	0.0161	0.0119	1.71	367.85	369.55	4.80	3.00
ktree_fr_k3	40	ARIEL	0.0034	0.0065	0.00	25.41	25.41	4.80	–
ktree_fr_k3	40	IBP	0.0029	0.0036	1.68	8.70	10.38	4.80	3.00
ktree_fr_k3	50	CDVE	0.0087	0.0133	3.68	845.10	848.78	6.89	3.00
ktree_fr_k3	50	ARIEL	0.0010	0.0077	0.00	31.32	31.32	6.89	–
ktree_fr_k3	50	IBP	0.0042	0.0030	3.64	11.44	15.08	6.89	3.00
ktree_fr_k3	100	ARIEL	0.0037	0.0090	0.00	64.03	64.03	22.71	–
ktree_fr_k3	100	IBP	0.0024	0.0048	10.55	25.34	35.89	22.71	3.00

Table 14: Mean error metrics vs. CJT reference

Type	n	Algorithm	MAE _{lb}	MAE _{ub}	Build	Run	Total	Ref	IW
dag_fr_tw3	5	CDVE	0.0039	0.0020	0.13	0.06	0.20	0.42	2.00
dag_fr_tw3	5	ARIEL	0.0090	0.0123	0.00	3.38	3.38	0.42	–
dag_fr_tw3	5	CVE	0.0006	0.0017	0.13	250.50	250.64	0.42	2.00
dag_fr_tw3	5	CVE _{cm}	0.0619	0.0578	0.13	0.03	0.16	0.42	2.00
dag_fr_tw3	5	IBP	0.0021	0.0020	0.13	0.01	0.13	0.42	2.00
dag_fr_tw3	6	CDVE	0.0031	0.0037	0.14	0.09	0.23	0.49	1.90
dag_fr_tw3	6	ARIEL	0.0008	0.0012	0.00	7.96	7.96	0.49	–
dag_fr_tw3	6	CVE	0.0031	0.0019	0.15	267.68	267.84	0.49	1.90
dag_fr_tw3	6	CVE _{cm}	0.0517	0.0569	0.14	0.04	0.18	0.49	1.90
dag_fr_tw3	6	IBP	0.0033	0.0020	0.13	0.01	0.14	0.49	1.90
dag_fr_tw3	7	CDVE	0.0076	0.0045	0.17	0.15	0.32	0.79	1.80
dag_fr_tw3	7	ARIEL	0.0000	0.0000	0.00	1.92	1.92	0.79	–
dag_fr_tw3	7	CVE	0.0031	0.0020	0.13	282.36	282.49	0.79	1.50
dag_fr_tw3	7	CVE _{cm}	0.0677	0.0778	0.17	0.07	0.23	0.79	1.80
dag_fr_tw3	7	IBP	0.0073	0.0041	0.16	0.01	0.17	0.79	1.80
dag_fr_tw3	8	CDVE	0.0015	0.0035	0.24	0.33	0.57	1.10	2.20
dag_fr_tw3	8	ARIEL	0.0029	0.0007	0.00	3.26	3.26	1.10	–
dag_fr_tw3	8	CVE _{cm}	0.1535	0.1293	0.24	0.14	0.39	1.10	2.20
dag_fr_tw3	8	IBP	0.0032	0.0021	0.24	0.02	0.26	1.10	2.20
dag_fr_tw3	9	CDVE	0.0013	0.0019	0.28	0.49	0.77	1.23	2.20
dag_fr_tw3	9	ARIEL	0.0089	0.0114	0.00	3.50	3.50	1.23	–
dag_fr_tw3	9	CVE _{cm}	0.1521	0.1462	0.27	0.22	0.49	1.23	2.20
dag_fr_tw3	9	IBP	0.0016	0.0004	0.28	0.02	0.30	1.23	2.20
dag_fr_tw3	10	CDVE	0.0021	0.0060	0.31	0.73	1.05	1.62	2.20
dag_fr_tw3	10	ARIEL	0.0029	0.0029	0.00	3.92	3.92	1.62	–
dag_fr_tw3	10	CVE	0.0000	0.0000	0.20	104.02	104.21	1.62	2.00
dag_fr_tw3	10	CVE _{cm}	0.1930	0.1972	0.30	0.30	0.60	1.62	2.20
dag_fr_tw3	10	IBP	0.0044	0.0058	0.31	0.03	0.33	1.62	2.20

Table 15: Standard deviation of error metrics vs. CJT reference

Type	n	Algorithm	SD _{lb}	SD _{ub}	Build	Run	Total	Ref	IW
dag_fr_tw3	5	CDVE	0.0238	0.0085	0.13	0.06	0.20	0.42	2.00
dag_fr_tw3	5	ARIEL	0.0258	0.0563	0.00	3.38	3.38	0.42	–
dag_fr_tw3	5	CVE	0.0041	0.0085	0.13	250.50	250.64	0.42	2.00
dag_fr_tw3	5	CVE _{cm}	0.0697	0.0798	0.13	0.03	0.16	0.42	2.00
dag_fr_tw3	5	IBP	0.0074	0.0083	0.13	0.01	0.13	0.42	2.00
dag_fr_tw3	6	CDVE	0.0168	0.0163	0.14	0.09	0.23	0.49	1.90
dag_fr_tw3	6	ARIEL	0.0047	0.0067	0.00	7.96	7.96	0.49	–
dag_fr_tw3	6	CVE	0.0168	0.0094	0.15	267.68	267.84	0.49	1.90
dag_fr_tw3	6	CVE _{cm}	0.0878	0.0830	0.14	0.04	0.18	0.49	1.90
dag_fr_tw3	6	IBP	0.0168	0.0092	0.13	0.01	0.14	0.49	1.90
dag_fr_tw3	7	CDVE	0.0205	0.0150	0.17	0.15	0.32	0.79	1.80
dag_fr_tw3	7	ARIEL	0.0000	0.0001	0.00	1.92	1.92	0.79	–
dag_fr_tw3	7	CVE	0.0160	0.0102	0.13	282.36	282.49	0.79	1.50
dag_fr_tw3	7	CVE _{cm}	0.1090	0.1145	0.17	0.07	0.23	0.79	1.80
dag_fr_tw3	7	IBP	0.0206	0.0143	0.16	0.01	0.17	0.79	1.80
dag_fr_tw3	8	CDVE	0.0061	0.0165	0.24	0.33	0.57	1.10	2.20
dag_fr_tw3	8	ARIEL	0.0136	0.0031	0.00	3.26	3.26	1.10	–
dag_fr_tw3	8	CVE _{cm}	0.1487	0.1343	0.24	0.14	0.39	1.10	2.20
dag_fr_tw3	8	IBP	0.0091	0.0096	0.24	0.02	0.26	1.10	2.20
dag_fr_tw3	9	CDVE	0.0109	0.0163	0.28	0.49	0.77	1.23	2.20
dag_fr_tw3	9	ARIEL	0.0834	0.0838	0.00	3.50	3.50	1.23	–
dag_fr_tw3	9	CVE _{cm}	0.1428	0.1327	0.27	0.22	0.49	1.23	2.20
dag_fr_tw3	9	IBP	0.0110	0.0018	0.28	0.02	0.30	1.23	2.20
dag_fr_tw3	10	CDVE	0.0101	0.0223	0.31	0.73	1.05	1.62	2.20
dag_fr_tw3	10	ARIEL	0.0211	0.0170	0.00	3.92	3.92	1.62	–
dag_fr_tw3	10	CVE	0.0000	0.0000	0.20	104.02	104.21	1.62	2.00
dag_fr_tw3	10	CVE _{cm}	0.1640	0.1650	0.30	0.30	0.60	1.62	2.20
dag_fr_tw3	10	IBP	0.0142	0.0190	0.31	0.03	0.33	1.62	2.20

Table 16: Mean error metrics vs. CJT reference

Type	n	Algorithm	MAE _{lb}	MAE _{ub}	Build	Run	Total	Ref	IW
dag_fr_tw3	10	CDVE	0.0012	0.0020	0.32	0.76	1.07	1.53	2.40
dag_fr_tw3	10	ARIEL	0.0013	0.0038	0.00	18.64	18.64	1.53	–
dag_fr_tw3	10	CVE _{cm}	0.1992	0.2057	0.32	0.41	0.72	1.53	2.40
dag_fr_tw3	10	IBP	0.0019	0.0030	0.31	0.03	0.34	1.53	2.40
dag_fr_tw3	20	CDVE	0.0039	0.0023	0.67	8.53	9.20	4.17	2.60
dag_fr_tw3	20	ARIEL	0.0040	0.0026	0.00	9.46	9.46	4.17	–
dag_fr_tw3	20	CVE _{cm}	0.3072	0.3114	0.65	23.43	24.08	4.17	2.60
dag_fr_tw3	20	IBP	0.0037	0.0026	0.66	0.07	0.73	4.17	2.60
dag_fr_tw3	30	CDVE	0.0025	0.0023	1.03	36.88	37.90	10.25	2.80
dag_fr_tw3	30	ARIEL	0.0013	0.0001	0.00	13.20	13.20	10.25	–
dag_fr_tw3	30	CVE _{cm}	0.3439	0.3465	0.95	284.01	284.96	10.25	2.75
dag_fr_tw3	30	IBP	0.0021	0.0018	0.94	0.09	1.03	10.25	2.80
dag_fr_tw3	40	CDVE	0.0039	0.0036	1.43	110.19	111.62	14.22	3.00
dag_fr_tw3	40	ARIEL	0.0009	0.0034	0.00	32.43	32.43	14.22	–
dag_fr_tw3	40	CVE _{cm}	0.3173	0.3677	1.09	1135.19	1136.29	14.22	3.00
dag_fr_tw3	40	IBP	0.0019	0.0018	1.35	0.17	1.51	14.22	3.00
dag_fr_tw3	50	CDVE	0.0037	0.0032	1.65	238.85	240.50	23.21	3.00
dag_fr_tw3	50	ARIEL	0.0011	0.0008	0.00	24.86	24.86	23.21	–
dag_fr_tw3	50	CVE _{cm}	0.3756	0.3915	1.54	194.40	195.94	23.21	3.00
dag_fr_tw3	50	IBP	0.0027	0.0020	1.67	0.20	1.87	23.21	3.00
dag_fr_tw3	100	CDVE	0.0030	0.0024	3.00	3278.99	3281.98	80.21	3.00
dag_fr_tw3	100	ARIEL	0.0013	0.0010	0.00	56.21	56.21	80.21	–
dag_fr_tw3	100	IBP	0.0013	0.0012	3.34	0.40	3.75	80.21	3.00

Table 17: Standard deviation of error metrics vs. CJT reference

Type	n	Algorithm	SD _{lb}	SD _{ub}	Build	Run	Total	Ref	IW
dag_fr_tw3	10	CDVE	0.0060	0.0079	0.32	0.76	1.07	1.53	2.40
dag_fr_tw3	10	ARIEL	0.0070	0.0196	0.00	18.64	18.64	1.53	–
dag_fr_tw3	10	CVE _{cm}	0.1675	0.1789	0.32	0.41	0.72	1.53	2.40
dag_fr_tw3	10	IBP	0.0066	0.0099	0.31	0.03	0.34	1.53	2.40
dag_fr_tw3	20	CDVE	0.0155	0.0093	0.67	8.53	9.20	4.17	2.60
dag_fr_tw3	20	ARIEL	0.0180	0.0147	0.00	9.46	9.46	4.17	–
dag_fr_tw3	20	CVE _{cm}	0.1853	0.1833	0.65	23.43	24.08	4.17	2.60
dag_fr_tw3	20	IBP	0.0115	0.0099	0.66	0.07	0.73	4.17	2.60
dag_fr_tw3	30	CDVE	0.0092	0.0122	1.03	36.88	37.90	10.25	2.80
dag_fr_tw3	30	ARIEL	0.0118	0.0022	0.00	13.20	13.20	10.25	–
dag_fr_tw3	30	CVE _{cm}	0.1779	0.1827	0.95	284.01	284.96	10.25	2.75
dag_fr_tw3	30	IBP	0.0085	0.0100	0.94	0.09	1.03	10.25	2.80
dag_fr_tw3	40	CDVE	0.0148	0.0189	1.43	110.19	111.62	14.22	3.00
dag_fr_tw3	40	ARIEL	0.0046	0.0312	0.00	32.43	32.43	14.22	–
dag_fr_tw3	40	CVE _{cm}	0.1591	0.2025	1.09	1135.19	1136.29	14.22	3.00
dag_fr_tw3	40	IBP	0.0081	0.0073	1.35	0.17	1.51	14.22	3.00
dag_fr_tw3	50	CDVE	0.0173	0.0148	1.65	238.85	240.50	23.21	3.00
dag_fr_tw3	50	ARIEL	0.0122	0.0063	0.00	24.86	24.86	23.21	–
dag_fr_tw3	50	CVE _{cm}	0.1807	0.1878	1.54	194.40	195.94	23.21	3.00
dag_fr_tw3	50	IBP	0.0124	0.0080	1.67	0.20	1.87	23.21	3.00
dag_fr_tw3	100	CDVE	0.0167	0.0142	3.00	3278.99	3281.98	80.21	3.00
dag_fr_tw3	100	ARIEL	0.0096	0.0083	0.00	56.21	56.21	80.21	–
dag_fr_tw3	100	IBP	0.0062	0.0064	3.34	0.40	3.75	80.21	3.00

2.3 Realistic LCNs

The two tables below give the complete per-instance results for the ten Bayesian-network-derived LCNs and the sixteen Supreme Court chain-graph LCNs, including the nine dense `supreme_court` instances omitted from Table 1. On those nine, only ARIEL and the CredalJT reference complete: with $n=10$ atoms but 345–1009 sentences, the size of the constraint system rather than the treewidth is the binding limit, and IBP, CDVE and all CVE variants exhaust the budget. One instance in the suite, `benchmark_scaling_directions_n24_2`, is a repaired variant of the original benchmark: in the chain graph the four reviewers of a stage are mutually independent, so the probability of their conjunction is forced to the product of their marginals, which the original’s `component` sentences contradicted; the three incompatible sentences were dropped and the repaired instance is consistent, verified by a junction-tree feasibility check.

Table 18: Mean error metrics vs. CJT reference

Instance	Algorithm	MAE _{ib}	MAE _{ub}	Build	Run	Total	Ref	IW
alarm_consistent.lcn	CDVE	0.1008	0.0146	0.10	0.59	0.69	0.72	2.00
alarm_consistent.lcn	ARIEL	0.0000	0.0000	0.00	3.92	3.92	0.72	–
alarm_consistent.lcn	CVE _{cm}	0.0866	0.0670	0.10	0.14	0.24	0.72	2.00
alarm_consistent.lcn	IBP	0.1008	0.0149	0.10	0.03	0.13	0.72	2.00
asia_consistent.lcn	CDVE	0.0033	0.0073	0.08	0.24	0.32	0.45	2.00
asia_consistent.lcn	ARIEL	0.0000	0.0000	0.00	2.86	2.86	0.45	–
asia_consistent.lcn	CVE _{cm}	0.0323	0.0291	0.08	0.08	0.16	0.45	2.00
asia_consistent.lcn	IBP	0.0035	0.0063	0.08	0.01	0.10	0.45	2.00
cancer_consistent.lcn	CDVE	0.0653	0.0272	0.06	0.06	0.12	0.37	2.00
cancer_consistent.lcn	ARIEL	0.0013	0.0139	0.00	2.10	2.10	0.37	–
cancer_consistent.lcn	CVE	0.0653	0.0272	0.06	42.46	42.52	0.37	2.00
cancer_consistent.lcn	CVE _{cm}	0.0491	0.0267	0.06	0.03	0.09	0.37	2.00
cancer_consistent.lcn	IBP	0.0639	0.0392	0.06	0.01	0.07	0.37	2.00
credit_consistent.lcn	CDVE	0.2570	0.0000	0.11	0.58	0.69	1.15	3.00
credit_consistent.lcn	ARIEL	0.0496	0.0000	0.00	3.18	3.18	1.15	–
credit_consistent.lcn	CVE _{cm}	0.1079	0.1650	0.11	1.62	1.73	1.15	3.00
credit_consistent.lcn	IBP	0.2570	0.0000	0.11	0.13	0.24	1.15	3.00
earthquake_consistent.lcn	CDVE	0.0041	0.0000	0.05	0.05	0.10	0.53	2.00
earthquake_consistent.lcn	ARIEL	0.0002	0.0000	0.00	1.88	1.88	0.53	–
earthquake_consistent.lcn	CVE	0.0041	0.0000	0.05	7.63	7.68	0.53	2.00
earthquake_consistent.lcn	CVE _{cm}	0.0019	0.0036	0.05	0.03	0.08	0.53	2.00
earthquake_consistent.lcn	IBP	0.0041	0.0000	0.05	0.01	0.06	0.53	2.00
engine_consistent.lcn	CDVE	0.0591	0.0104	0.10	0.43	0.53	0.53	2.00
engine_consistent.lcn	ARIEL	0.0000	0.0000	0.00	3.25	3.25	0.53	–
engine_consistent.lcn	CVE _{cm}	0.1234	0.0839	0.10	0.32	0.42	0.53	2.00
engine_consistent.lcn	IBP	0.0591	0.0104	0.10	0.01	0.12	0.53	2.00
hepatitis_consistent.lcn	CDVE	0.0703	0.0851	0.24	0.90	1.15	0.56	3.00
hepatitis_consistent.lcn	ARIEL	0.0084	0.0075	0.00	3.91	3.91	0.56	–
hepatitis_consistent.lcn	CVE _{cm}	0.1137	0.1838	0.24	16.29	16.53	0.56	3.00
hepatitis_consistent.lcn	IBP	0.0686	0.0797	0.24	0.50	0.75	0.56	3.00
suicide_consistent.lcn	CDVE	0.0849	0.0302	0.10	0.52	0.62	0.87	2.00
suicide_consistent.lcn	ARIEL	0.0000	0.0000	0.00	3.74	3.74	0.87	–
suicide_consistent.lcn	CVE _{cm}	0.0970	0.0513	0.10	0.13	0.23	0.87	2.00
suicide_consistent.lcn	IBP	0.0849	0.0302	0.10	0.02	0.12	0.87	2.00
tank_consistent.lcn	CDVE	0.1149	0.0751	0.30	0.87	1.18	0.77	3.00
tank_consistent.lcn	ARIEL	0.0048	0.0034	0.00	5.02	5.02	0.77	–
tank_consistent.lcn	CVE _{cm}	0.0653	0.0802	0.31	0.19	0.50	0.77	3.00
tank_consistent.lcn	IBP	0.1163	0.0739	0.30	0.02	0.32	0.77	3.00
toy_consistent.lcn	CDVE	0.0891	0.0745	0.09	0.03	0.12	0.16	2.00
toy_consistent.lcn	ARIEL	0.0000	0.0000	0.00	1.60	1.60	0.16	–
toy_consistent.lcn	CVE _{cm}	0.0742	0.0294	0.09	0.01	0.10	0.16	2.00
toy_consistent.lcn	IBP	0.0891	0.0745	0.09	0.01	0.09	0.16	2.00

Table 19: Standard deviation of error metrics vs. CJT reference

Instance	Algorithm	SD _{lb}	SD _{ub}	Build	Run	Total	Ref	IW
alarm_consistent.lcn	CDVE	0.1476	0.0203	0.10	0.59	0.69	0.72	2.00
alarm_consistent.lcn	ARIEL	0.0000	0.0000	0.00	3.92	3.92	0.72	–
alarm_consistent.lcn	CVE _{cm}	0.1279	0.0854	0.10	0.14	0.24	0.72	2.00
alarm_consistent.lcn	IBP	0.1476	0.0208	0.10	0.03	0.13	0.72	2.00
asia_consistent.lcn	CDVE	0.0043	0.0172	0.08	0.24	0.32	0.45	2.00
asia_consistent.lcn	ARIEL	0.0000	0.0000	0.00	2.86	2.86	0.45	–
asia_consistent.lcn	CVE _{cm}	0.0308	0.0359	0.08	0.08	0.16	0.45	2.00
asia_consistent.lcn	IBP	0.0047	0.0144	0.08	0.01	0.10	0.45	2.00
cancer_consistent.lcn	CDVE	0.0374	0.0320	0.06	0.06	0.12	0.37	2.00
cancer_consistent.lcn	ARIEL	0.0026	0.0278	0.00	2.10	2.10	0.37	–
cancer_consistent.lcn	CVE	0.0374	0.0320	0.06	42.46	42.52	0.37	2.00
cancer_consistent.lcn	CVE _{cm}	0.0397	0.0233	0.06	0.03	0.09	0.37	2.00
cancer_consistent.lcn	IBP	0.0369	0.0323	0.06	0.01	0.07	0.37	2.00
credit_consistent.lcn	CDVE	0.2149	0.0000	0.11	0.58	0.69	1.15	3.00
credit_consistent.lcn	ARIEL	0.0414	0.0000	0.00	3.18	3.18	1.15	–
credit_consistent.lcn	CVE _{cm}	0.1401	0.1170	0.11	1.62	1.73	1.15	3.00
credit_consistent.lcn	IBP	0.2149	0.0000	0.11	0.13	0.24	1.15	3.00
earthquake_consistent.lcn	CDVE	0.0035	0.0000	0.05	0.05	0.10	0.53	2.00
earthquake_consistent.lcn	ARIEL	0.0003	0.0000	0.00	1.88	1.88	0.53	–
earthquake_consistent.lcn	CVE	0.0035	0.0000	0.05	7.63	7.68	0.53	2.00
earthquake_consistent.lcn	CVE _{cm}	0.0029	0.0043	0.05	0.03	0.08	0.53	2.00
earthquake_consistent.lcn	IBP	0.0035	0.0000	0.05	0.01	0.06	0.53	2.00
engine_consistent.lcn	CDVE	0.1131	0.0131	0.10	0.43	0.53	0.53	2.00
engine_consistent.lcn	ARIEL	0.0000	0.0000	0.00	3.25	3.25	0.53	–
engine_consistent.lcn	CVE _{cm}	0.1318	0.0677	0.10	0.32	0.42	0.53	2.00
engine_consistent.lcn	IBP	0.1131	0.0131	0.10	0.01	0.12	0.53	2.00
hepatitis_consistent.lcn	CDVE	0.1232	0.1429	0.24	0.90	1.15	0.56	3.00
hepatitis_consistent.lcn	ARIEL	0.0252	0.0167	0.00	3.91	3.91	0.56	–
hepatitis_consistent.lcn	CVE _{cm}	0.1580	0.1630	0.24	16.29	16.53	0.56	3.00
hepatitis_consistent.lcn	IBP	0.1236	0.1445	0.24	0.50	0.75	0.56	3.00
suicide_consistent.lcn	CDVE	0.1160	0.0430	0.10	0.52	0.62	0.87	2.00
suicide_consistent.lcn	ARIEL	0.0000	0.0000	0.00	3.74	3.74	0.87	–
suicide_consistent.lcn	CVE _{cm}	0.1147	0.0637	0.10	0.13	0.23	0.87	2.00
suicide_consistent.lcn	IBP	0.1160	0.0430	0.10	0.02	0.12	0.87	2.00
tank_consistent.lcn	CDVE	0.1337	0.1096	0.30	0.87	1.18	0.77	3.00
tank_consistent.lcn	ARIEL	0.0144	0.0102	0.00	5.02	5.02	0.77	–
tank_consistent.lcn	CVE _{cm}	0.0659	0.0555	0.31	0.19	0.50	0.77	3.00
tank_consistent.lcn	IBP	0.1351	0.1089	0.30	0.02	0.32	0.77	3.00
toy_consistent.lcn	CDVE	0.1363	0.0751	0.09	0.03	0.12	0.16	2.00
toy_consistent.lcn	ARIEL	0.0000	0.0000	0.00	1.60	1.60	0.16	–
toy_consistent.lcn	CVE _{cm}	0.1190	0.0345	0.09	0.01	0.10	0.16	2.00
toy_consistent.lcn	IBP	0.1363	0.0751	0.09	0.01	0.09	0.16	2.00

Table 20: Mean error metrics vs. CJT reference

Instance	Algorithm	MAE _{lb}	MAE _{ub}	Build	Run	Total	Ref	IW
benchmark_scaling_directions_n24_2.lcn	CDVE	0.1215	0.4370	1.78	22.80	24.57	1258.24	4.00
benchmark_scaling_directions_n24_2.lcn	ARIEL	0.0000	0.0000	0.00	13.42	13.42	1258.24	–
benchmark_scaling_directions_n24_2.lcn	CVE _{cm}	0.1123	0.1749	1.86	2206.67	2208.54	1258.24	4.00
benchmark_scaling_directions_n24_2.lcn	IBP	0.1215	0.4370	1.91	685.00	686.91	1258.24	4.00
benchmark_scaling_directions_n32_1.lcn	CDVE	0.1148	0.2247	0.57	19.98	20.55	2.42	1.00
benchmark_scaling_directions_n32_1.lcn	ARIEL	0.0000	0.0000	0.00	19.80	19.80	2.42	–
benchmark_scaling_directions_n32_1.lcn	CVE _{cm}	0.0643	0.0716	0.59	8.59	9.18	2.42	1.00
benchmark_scaling_directions_n32_1.lcn	IBP	0.1148	0.2247	0.64	0.05	0.69	2.42	1.00
maintenance_scaling_n12_3.lcn	CDVE	0.1287	0.5013	0.29	0.94	1.23	0.66	2.00
maintenance_scaling_n12_3.lcn	ARIEL	0.0284	0.0000	0.00	8.13	8.13	0.66	–
maintenance_scaling_n12_3.lcn	CVE _{cm}	0.0793	0.2270	0.31	0.25	0.57	0.66	2.00
maintenance_scaling_n12_3.lcn	IBP	0.1286	0.4926	0.31	0.04	0.35	0.66	2.00
maintenance_scaling_n6_1.lcn	CDVE	0.1029	0.3972	0.17	0.10	0.27	0.28	2.00
maintenance_scaling_n6_1.lcn	ARIEL	0.0000	0.0000	0.00	4.16	4.16	0.28	–
maintenance_scaling_n6_1.lcn	CVE	0.1029	0.4017	0.14	789.77	789.91	0.28	2.00
maintenance_scaling_n6_1.lcn	CVE _{cm}	0.1021	0.2685	0.14	0.05	0.19	0.28	2.00
maintenance_scaling_n6_1.lcn	IBP	0.1029	0.4017	0.15	0.01	0.16	0.28	2.00
maintenance_scaling_n9_2.lcn	CDVE	0.1447	0.4748	0.21	0.48	0.69	0.44	2.00
maintenance_scaling_n9_2.lcn	ARIEL	0.0398	0.0000	0.00	6.09	6.09	0.44	–
maintenance_scaling_n9_2.lcn	CVE _{cm}	0.1095	0.2437	0.20	0.12	0.32	0.44	2.00
maintenance_scaling_n9_2.lcn	IBP	0.1447	0.4662	0.23	0.02	0.25	0.44	2.00
paper_scaling_directions_n35_2.lcn	CDVE	0.1506	0.4467	7.59	190.48	198.07	1342.14	5.00
paper_scaling_directions_n35_2.lcn	ARIEL	0.0064	0.0047	0.00	36.58	36.58	1342.14	–
paper_scaling_directions_n40_1.lcn	CDVE	0.1288	0.2631	0.70	50.55	51.25	3.27	1.00
paper_scaling_directions_n40_1.lcn	ARIEL	0.0000	0.0000	0.00	34.72	34.72	3.27	–
paper_scaling_directions_n40_1.lcn	CVE _{cm}	0.0696	0.0566	0.74	658.10	658.84	3.27	1.00
paper_scaling_directions_n40_1.lcn	IBP	0.1288	0.2631	0.82	0.09	0.91	3.27	1.00
supreme_court_judicial_power_credal_source_comparison_n10_1.lcn	ARIEL	0.0532	0.1234	0.00	115.72	115.72	348.07	–
supreme_court_judicial_power_credal_source_comparison_n10_2.lcn	ARIEL	0.0321	0.1107	0.00	76.72	76.72	337.23	–
supreme_court_judicial_power_credal_source_comparison_n10_3.lcn	ARIEL	0.0259	0.0972	0.00	527.92	527.92	426.93	–
supreme_court_judicial_power_dual_source_credal_n10_1.lcn	ARIEL	0.0259	0.0972	0.00	523.64	523.64	435.71	–
supreme_court_judicial_power_release_comparison_n10_1.lcn	ARIEL	0.0317	0.0860	0.00	68.03	68.03	346.18	–
supreme_court_judicial_power_release_comparison_n10_2.lcn	ARIEL	0.0532	0.1234	0.00	117.31	117.31	337.95	–
supreme_court_judicial_power_release_comparison_n10_3.lcn	ARIEL	0.0259	0.0972	0.00	562.73	562.73	426.77	–
supreme_court_judicial_power_source_variants_n10_1.lcn	ARIEL	0.0317	0.0860	0.00	71.64	71.64	338.65	–
supreme_court_judicial_power_source_variants_n10_2.lcn	ARIEL	0.0176	0.0913	0.00	70.58	70.58	357.82	–

References

- [1] Anthony Constantinou, Yang Liu, Kiattikun Chobtham, Zhigao Guo, and Neville Kitson. The bayesys data and bayesian network repository. Technical report, Bayesian Artificial Intelligence research lab, Queen Mary University of London, London, UK, 2020.
- [2] Elizabeth L. Ogburn, Ilya Shpitser, and Youjin Lee. Causal inference, social networks, and chain graphs, 2020.
- [3] Radu Marinescu, Haifeng Qian, Alexander Gray, Debarun Bhattacharjya, Francisco Barahona, Tian Gao, and Ryan Riegel. Approximate inference in logical credal networks. In *32nd International Joint Conference on Artificial Intelligence (IJCAI)*, 2023.
- [4] Andreas Wächter and Lorenz Biegler. On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming. *Mathematical Programming*, 106(1):25–57, 2006.

Table 21: Standard deviation of error metrics vs. CJT reference

Instance	Algorithm	SD _{lb}	SD _{ub}	Build	Run	Total	Ref	IW
benchmark_scaling_directions_n24_2.lcn	CDVE	0.0648	0.1928	1.78	22.80	24.57	1258.24	4.00
benchmark_scaling_directions_n24_2.lcn	ARIEL	0.0000	0.0000	0.00	13.42	13.42	1258.24	–
benchmark_scaling_directions_n24_2.lcn	CVE _{cm}	0.0780	0.1686	1.86	2206.67	2208.54	1258.24	4.00
benchmark_scaling_directions_n24_2.lcn	IBP	0.0648	0.1928	1.91	685.00	686.91	1258.24	4.00
benchmark_scaling_directions_n32_1.lcn	CDVE	0.0657	0.1463	0.57	19.98	20.55	2.42	1.00
benchmark_scaling_directions_n32_1.lcn	ARIEL	0.0000	0.0000	0.00	19.80	19.80	2.42	–
benchmark_scaling_directions_n32_1.lcn	CVE _{cm}	0.0430	0.0738	0.59	8.59	9.18	2.42	1.00
benchmark_scaling_directions_n32_1.lcn	IBP	0.0657	0.1463	0.64	0.05	0.69	2.42	1.00
maintenance_scaling_n12_3.lcn	CDVE	0.0795	0.2363	0.29	0.94	1.23	0.66	2.00
maintenance_scaling_n12_3.lcn	ARIEL	0.0514	0.0000	0.00	8.13	8.13	0.66	–
maintenance_scaling_n12_3.lcn	CVE _{cm}	0.0491	0.1572	0.31	0.25	0.57	0.66	2.00
maintenance_scaling_n12_3.lcn	IBP	0.0794	0.2313	0.31	0.04	0.35	0.66	2.00
maintenance_scaling_n6_1.lcn	CDVE	0.0729	0.2197	0.17	0.10	0.27	0.28	2.00
maintenance_scaling_n6_1.lcn	ARIEL	0.0000	0.0000	0.00	4.16	4.16	0.28	–
maintenance_scaling_n6_1.lcn	CVE	0.0729	0.2232	0.14	789.77	789.91	0.28	2.00
maintenance_scaling_n6_1.lcn	CVE _{cm}	0.0710	0.1599	0.14	0.05	0.19	0.28	2.00
maintenance_scaling_n6_1.lcn	IBP	0.0729	0.2232	0.15	0.01	0.16	0.28	2.00
maintenance_scaling_n9_2.lcn	CDVE	0.0953	0.2482	0.21	0.48	0.69	0.44	2.00
maintenance_scaling_n9_2.lcn	ARIEL	0.0496	0.0000	0.00	6.09	6.09	0.44	–
maintenance_scaling_n9_2.lcn	CVE _{cm}	0.0699	0.1858	0.20	0.12	0.32	0.44	2.00
maintenance_scaling_n9_2.lcn	IBP	0.0952	0.2428	0.23	0.02	0.25	0.44	2.00
paper_scaling_directions_n35_2.lcn	CDVE	0.0682	0.1408	7.59	190.48	198.07	1342.14	5.00
paper_scaling_directions_n35_2.lcn	ARIEL	0.0250	0.0276	0.00	36.58	36.58	1342.14	–
paper_scaling_directions_n40_1.lcn	CDVE	0.0656	0.1334	0.70	50.55	51.25	3.27	1.00
paper_scaling_directions_n40_1.lcn	ARIEL	0.0000	0.0000	0.00	34.72	34.72	3.27	–
paper_scaling_directions_n40_1.lcn	CVE _{cm}	0.0672	0.0365	0.74	658.10	658.84	3.27	1.00
paper_scaling_directions_n40_1.lcn	IBP	0.0656	0.1334	0.82	0.09	0.91	3.27	1.00
supreme_court_judicial_power_credal_source_comparison_n10_1.lcn	ARIEL	0.0620	0.2451	0.00	115.72	115.72	348.07	–
supreme_court_judicial_power_credal_source_comparison_n10_2.lcn	ARIEL	0.0407	0.2477	0.00	76.72	76.72	337.23	–
supreme_court_judicial_power_credal_source_comparison_n10_3.lcn	ARIEL	0.0414	0.2514	0.00	527.92	527.92	426.93	–
supreme_court_judicial_power_dual_source_credal_n10_1.lcn	ARIEL	0.0414	0.2514	0.00	523.64	523.64	435.71	–
supreme_court_judicial_power_release_comparison_n10_1.lcn	ARIEL	0.0624	0.2550	0.00	68.03	68.03	346.18	–
supreme_court_judicial_power_release_comparison_n10_2.lcn	ARIEL	0.0620	0.2451	0.00	117.31	117.31	337.95	–
supreme_court_judicial_power_release_comparison_n10_3.lcn	ARIEL	0.0414	0.2514	0.00	562.73	562.73	426.77	–
supreme_court_judicial_power_source_variants_n10_1.lcn	ARIEL	0.0624	0.2550	0.00	71.64	71.64	338.65	–
supreme_court_judicial_power_source_variants_n10_2.lcn	ARIEL	0.0442	0.2536	0.00	70.58	70.58	357.82	–